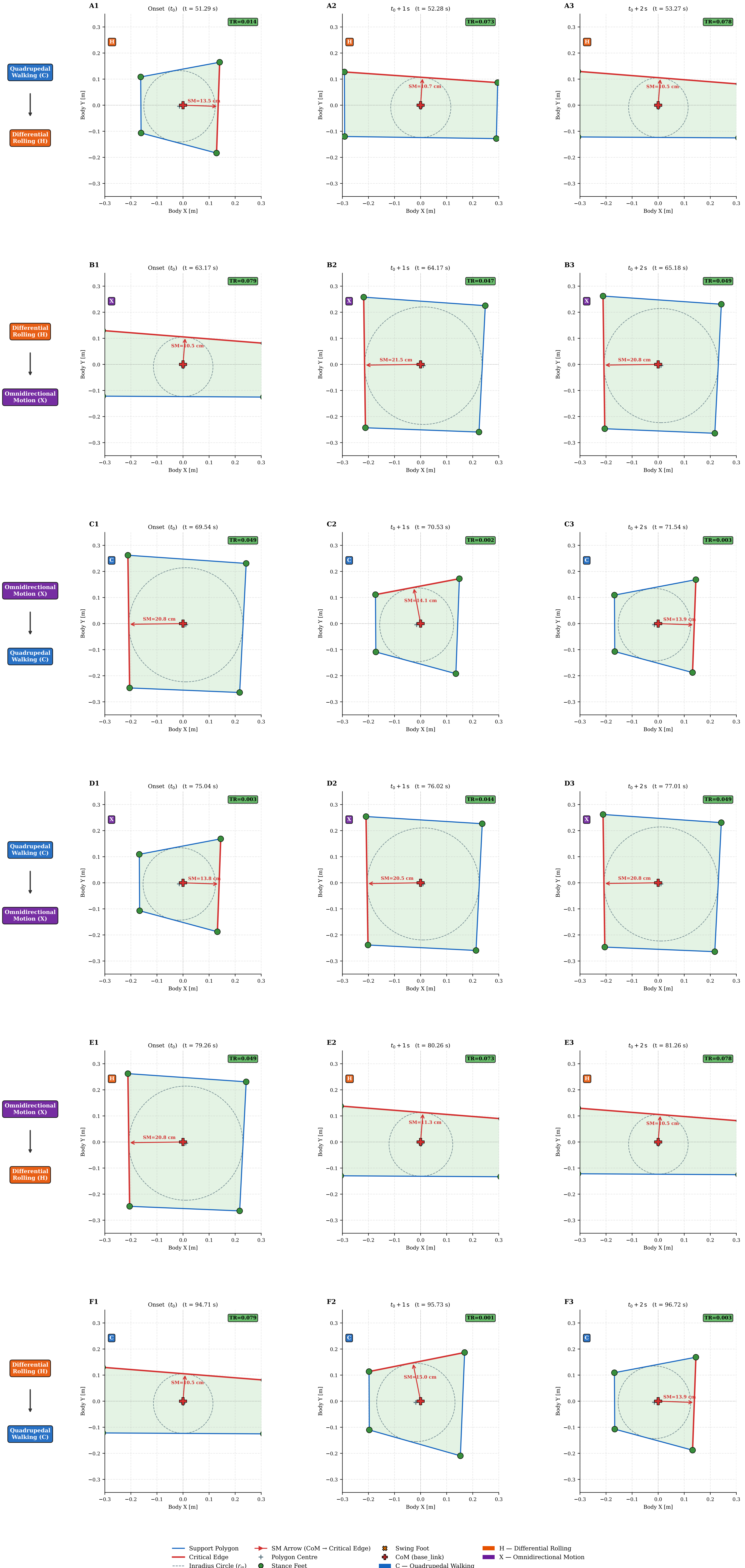


Static Stability Analysis across Locomotion Mode Transitions Generalised McGhee-Frank (1968) Support Polygon



SM=13.5 cm

SM=10.7 cm

SM=10.5 cm

SM=10.5 cm

SM=21.5 cm

SM=20.8 cm

SM=20.8 cm

SM=14.1 cm

SM=13.9 cm

SM=13.8 cm

SM=20.5 cm

SM=20.8 cm

SM=20.8 cm

SM=11.3 cm

SM=10.5 cm

SM=10.5 cm

SM=15.0 cm

SM=13.9 cm

Support Polygon

Critical Edge

Inradius Circle (r_m)

Stance Feet

SM Arrow (CoM → Critical Edge)

Polygon Centre

CoM (base_link)

Swing Foot

C — Quadrupedal Walking

H — Differential Rolling

X — Omnidirectional Motion